

Shay Mordechai

Security Researcher | Vulnerability Research, Reverse Engineering & OS Internals

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Summary

Offensive Security Researcher specializing in vulnerability discovery, Reverse Engineering, and OS Internals. Proven expertise in bypassing anti-analysis mechanisms, auditing complex execution environments (JIT, Compilers), and authoring exploit reports.

Professional Experience

Security Architect & Researcher | MyLeads AI (Independent) | Jan 2026 – Present

- **Offensive Auditing:** Mapped and exploited API architectures using Burp Suite, mitigating critical IDORs and logic flaws.
- **Rust/Wasm DLP Firewall:** Engineered a WebAssembly filter for Envoy Proxy to dynamically redact sensitive egress traffic.
- **OS Isolation:** Hardened execution environments bypassing virtualization using Rootless Podman and Linux namespaces.

DevSecOps Intern | Israel Tax Authority | Aug 2024 – Jul 2025

- **AppSec & CI/CD:** Integrated Checkmarx SAST/SCA into 15+ pipelines. Triaged findings and trained R&D on OWASP Top 10.
- **Defense-in-Depth:** Managed Threat Modeling and hardened IBM DataPower gateways via strict JSON validation schemas.

Combat Soldier | Intelligence Collection Corps, IDF (Hesder) | Aug 2016 – Apr 2018

- Executed covert reconnaissance operations, deploying stealth observation posts for operational target profiling.

Education

B.Sc. Computer Science | Jerusalem College of Technology | 2021 – 2025 (GPA: 89)

- **Relevant Coursework:** Reverse Engineering (93), Network Analysis (97), Information Security (90).

Technical Projects & Research *(Detailed Write-ups at shay-mordechai.github.io)*

- **Advanced Reverse Engineering Challenges:** Solved a series of highly selective challenges (including the ISA/Shabak Level 1 challenge). Demonstrated expertise in analyzing multi-stage malware, bypassing TLS Callbacks and anti-debugging checks, executing Entry Point Hijacking, Code Cave injections, and extracting hidden PE payloads via custom C++ decryptors.
- **AI-Augmented Vuln Research (SBCL Compiler):** Discovered 3 pending Zero-Days (CERT-IL #11265) via semantic fuzzing.
- **Network Exfiltration & C2 Simulation:** Engineered a CTF platform modeling nation-state tactics, implementing ICMP covert channels, PKI MITM, and steganography.
- **TLS 1.2 Cryptographic Simulator:** Authored a 4,000+ line Python/Scapy framework to simulate full TLS 1.2 handshakes from scratch, including automated SSLKeyLog generation for DPI.
- **V8 JIT Internals:** Published technical analysis on RWX memory allocation patterns and CFI bypasses in browser engines.

Technical Skills

- **Research & RE:** Reverse Engineering, Malware Analysis, Binary Patching, Fuzzing, Hooking, Code Injection.
- **Tools:** IDA Pro, x64dbg, WinDbg, Wireshark, Burp Suite, Sysinternals.
- **Languages:** C/C++, x86 Assembly, Python, Rust, Bash.
- **Specialization:** OS Internals (Linux/Windows), PE Format, L2-L7 Protocols, V8 JIT.